

Pandas Lab Program — Q39 & Q40

Question 39: Get the First 3 Rows of a DataFrame

Aim

To write a Pandas program to get the first three rows of a given DataFrame.

Source Code

```
import pandas as pd

data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily'],
    'score': [12.5, 9, 16.5, 20, 14],
    'attempts': [1, 3, 2, 3, 2],
    'qualify': ['yes', 'no', 'yes', 'yes', 'no']
}
labels = ['a', 'b', 'c', 'd', 'e']

df = pd.DataFrame(data, index=labels)
print("First 3 rows of the DataFrame:\n")
print(df.head(3))
# Alternative: print(df.iloc[:3])
```

Output

First 3 rows of the DataFrame:

	name	score	attempts	qualify
a	Anastasia	12.5	1	yes
b	Dima	9.0	3	no
c	Katherine	16.5	2	yes

Result

The Python program was executed successfully and the first three rows of the DataFrame (index labels 'a', 'b', 'c') were retrieved and displayed using the head() method.

Question 40: Select 'name' and 'score' Columns from a DataFrame

Aim

To write a Pandas program to select the 'name' and 'score' columns from the given DataFrame.

Source Code

```
import pandas as pd

data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily'],
    'score': [12.5, 9, 16.5, 20, 14],
    'attempts': [1, 3, 2, 3, 2],
    'qualify': ['yes', 'no', 'yes', 'yes', 'no']
}
labels = ['a', 'b', 'c', 'd', 'e']

df = pd.DataFrame(data, index=labels)
print("Selected 'name' and 'score' columns:\n")
print(df[['name', 'score']])
```

Output

Selected 'name' and 'score' columns:

	name	score
a	Anastasia	12.5
b	Dima	9.0
c	Katherine	16.5
d	James	20.0
e	Emily	14.0

Result

The Python program was executed successfully and only the 'name' and 'score' columns of the DataFrame were selected and displayed for all five records.